

ECO102: Introduction to Macroeconomics

Lecture 15: Money, The Price Level, and Inflation

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What is Money

- Any commodity or token that is generally acceptable as a means of payment is defined as money
- A ***means of payment*** is a method of settling a debt
- ***Wampum, tobacco, and nickels and dimes***
- Money serves three other functions:
 - ***Medium of exchange***: A medium of exchange is any object that is generally accepted in exchange for goods and services
 - ***Unit of account***: A unit of account is an agreed measure for stating the prices of goods and services
 - ***Store of value***: Money is a store of value in the sense that it can be held and exchanged later for goods and services

Measurements of Money

- In most countries money primarily consists of:
- **Currency:** The notes and coins held by individuals and businesses are known as currency
- **Deposits:** Deposits of individuals and businesses at banks and other depository institutions, such as savings and loan associations, are also counted as money
- **Other measures of Money:** Two official measures of money are known as M1 and M2.

M1 consists of currency and traveler's checks plus checking deposits owned by individuals and businesses. M1 does not include currency held by banks, and it does not include currency and checking deposits owned by the government

M2 consists of M1 plus time deposits, savings deposits, and money market mutual funds and other deposits

Measurements of Money

- Money is the means of payment, so the test of whether an asset is money is whether it serves as a means of payment
- Both currency and checking deposits are money because they can be transferred from one person to another by writing a check or using a debit card; meaning M1 is money
- Some of the savings deposits in M2 are just as much a means of payment as the checking deposits in M1
- Some savings deposits are not means of payment, these deposits are known as liquid assets. **Liquidity** is the property of being easily convertible into a means of payment without loss in value
- Since the deposits in M2 can quickly be converted into a means of payment they are counted as money
 - ***ARE DEPOSITS, CHECKS and CREDIT CARDS MONEY?***

Depository Institutions

- A depository institution is a financial firm that takes deposits from households and firms. These deposits are components of M1 and M2
- The deposits of three types of financial firms make up the nation's money:
- **Commercial Banks:** A commercial bank is a firm that is licensed to receive deposits and make loans.
- **Thrift Institutions:** Savings and loan associations, savings banks, and credit unions are thrift institutions
 - **Savings and Loan Association:** It is a is a depository institution that receives deposits and makes personal, commercial, and home purchase loans
 - **Savings Bank:** It is a is a depository institution that accepts savings deposits and makes mostly home-purchase loans
 - **Credit Union:** It is a is a depository institution that is owned by a social or economic group, such as a firm's employees, that accepts savings deposits and makes mostly personal loans

Depository Institutions

- ***Money Market Mutual Funds:*** A money market mutual fund is a fund operated by a financial institution that sells shares in the fund and holds assets such as U.S. Treasury bills and short-term commercial bills

Money market mutual fund shares act like bank deposits. Shareholders can write checks on their money market mutual fund accounts, but there are restrictions on most of these accounts

Money market mutual funds do not feature in M1 but counted in M2

Functions and Benefits of Depository Institutions

- Depository institutions provide services such as check clearing, account management, credit cards, and Internet banking, all of which provide an income from service fees
- But depository institutions earn most of their income by using the funds they receive from depositors to make loans and to buy securities that earn a higher interest rate than that paid to depositors
- A commercial bank puts the funds it receives from depositors and other funds that it borrows into four types of assets:
 - ***Reserves***
 - ***Liquid Assets***
 - ***Securities***
 - ***Loans***

Functions and Benefits of Depository Institutions

- **Reserves:** A bank's reserves are notes and coins in the bank's vault or in a deposit account at the Federal Reserve
- **Liquid Assets:** These are overnight loans to other banks, U.S. government Treasury bills, and commercial bills. These assets are the banks' first line of defense if they need reserves

The interest rate on overnight loans to other banks, called the **federal funds rate**, is targeted by the Fed
- **Securities:** These are U.S. government bonds and other bonds such as mortgage-backed securities. These assets can be sold and converted into reserves but at prices that fluctuate, so they are riskier than liquid assets and have a higher interest rate
- **Loans:** These are funds committed for an agreed-upon period of time to corporations to finance investment and to households to finance the purchase of homes, cars, and other durable goods. The outstanding balances on credit card accounts are also bank loans.

Loans are a bank's riskiest and highest-earning assets

Functions and Benefits of Depository Institutions

- Depository institutions provide four benefits:
- **Create Liquidity:** Depository institutions create liquidity by borrowing short and lending long
- **Pool risk:** Depository Institutions pool risk by lending out to a lot of people
- **Lower the cost of borrowing:** Depository institutions lower the cost of this search by providing large amounts of loans
- **Lower the cost of monitoring borrowers:** By monitoring borrowers, a lender can encourage good decisions that prevent defaults. But this activity is costly. Depository institutions can perform this task at a much lower cost
- Depository institutions are engaged in a risky business, and a failure, especially of a large bank, would have damaging effects on the entire financial system and economy. To make the risk of failure small, depository institutions are required to hold levels of reserves and owners' capital that equal or surpass ratios laid down by regulation.

Financial Innovation

- In the pursuit of larger profit, depository institutions are constantly seeking ways to improve their products in a process called ***financial innovation***
- The development of low-cost computing and communication brought financial innovations such as credit cards and daily interest deposit accounts
- Financial innovation has brought changes in the composition of money.
- Checking deposits at thrift institutions have become an increasing percentage of M1 while checking deposits at commercial banks have become a decreasing percentage
- Savings deposits have decreased as a percentage of M2, while time deposits and money market mutual funds have expanded.
- Surprisingly, the use of currency has not fallen much

The Federal Reserve System

- The Federal Reserve System (usually called the Fed) is the central bank of the United States. A **central bank** is a bank's bank and a public authority that regulates a nation's depository institutions and conducts monetary policy, which means that it adjusts the quantity of money in circulation and influences interest rates.
- Three key elements of the Fed's structure are:
- **The Board of Governors:** A seven-member board appointed by the President of the United States and confirmed by the Senate governs the Fed
- **The regional Federal Reserve banks:** The nation is divided into 12 Federal Reserve districts consisting of a Federal Reserve Bank that provides check-clearing services to commercial banks and issues bank notes
- **The Federal Open Market Committee:** The FOMC is the main policymaking organ of the Federal Reserve System that meets approximately every six weeks to review the state of the economy and to decide the actions to be carried out by the New York Fed

The Fed's Balance Sheet

- The Fed influences the economy through the size and composition of its balance sheet; the assets that the Fed owns and the liabilities that it owes
- ***The Fed's Assets:*** The Fed has two main assets
 - ***U.S. Government Securities:*** The Fed holds U.S. securities, Treasury bills and Treasury bonds, that it buys in the bond market. When the Fed buys or sells bonds, it participates in the loanable funds market
 - ***Loans to depository institutions:*** When depository institutions in aggregate are short of reserves, they can borrow from the Fed
- ***The Fed's Liabilities:*** The Fed has two main liabilities
 - ***Federal Reserve notes:*** Federal Reserve notes are the dollar bills that we use in our daily transactions
 - ***Depository institution deposits:*** Depository institution deposits at the Fed are part of the reserves of these institutions
- The ***monetary base*** is the sum of currency (Federal Reserve notes and coins issued by the treasury) and depository institution deposits at the Fed